

AI Assistance / Prompt Log

Student Details

Name: W.W.D.M. Fernando

Registration Number: IT23629844

Module: IE3010 - Network Programming

Project: *NetMessenger: A Multi-Client Chat and File-Sharing Platform over TCP/IP*

Purpose of AI Use

During the development of **NetMessenger**, I used ChatGPT mainly as a technical support and reference tool while developing and testing the application.

I used ChatGPT to:

- understand the assignment requirements;
- calculate the registration-specific values;
- understand TCP socket programming;
- troubleshoot compilation and runtime errors;
- understand TCP message framing;
- implement multiple-client communication;
- understand registration and presence management;
- implement broadcast and private messaging;
- implement chat-room functionality;
- implement file transfer;
- understand server-side file storage;
- implement error handling and graceful disconnection;
- review logging requirements;
- plan testing procedures;
- troubleshoot Git and GitHub issues;
- prepare documentation and report evidence.

I used the explanations as guidance and then **compiled, executed and tested the application in my own CentOS Linux environment.**

1. Assignment Requirements

Date-October 2026

Tool-ChatGPT

Prompt / Question-Asked for help understanding the main requirements of the IE3010 Network Programming assignment and identifying the mandatory functionality required for the NetMessenger application.

Support Received

ChatGPT helped me break the assignment into the following implementation areas:

- user registration;
- multiple simultaneous clients;
- user listing;
- broadcast messaging;
- private messaging;
- chat rooms;
- room messaging;
- file sharing;
- error handling;
- graceful client disconnection;
- server-side logging;
- personalized filenames, port number and Node ID.

How I Used the Response

I used this information to organize the development into smaller implementation and testing stages. I then implemented and tested the functionality in my Linux environment.

2. Personalized Assignment Values

Date-October 2026

Tool-ChatGPT

Prompt / Question-Asked how the personalized values should be calculated using my registration number: **-IT23629844**

Support Received-The personalized values were determined as follows:

Item	Value
Registration Number	IT23629844
Last 4 digits	9844
Server Port	15844
Node ID	NID:6298
Server Source	server_9844.c
Client Source	client_9844.c
Makefile	Makefile_9844
Log File	netmsg_IT23629844.log
Storage Path	./storage/IT23629844/<sender_username>/<filename>

How I Used the Response

I checked these values against the assignment requirements and used them consistently in the source files, Makefile, log file and storage structure.

3. Basic TCP Socket Programming

Tool-ChatGPT

Prompt / Question-Asked for an explanation of the normal process for implementing a TCP server and TCP client in C using BSD sockets.

Support Received-ChatGPT explained the purpose of functions including:

- socket()
- bind()
- listen()
- accept()
- connect()
- send()
- recv()
- close()

How I Used the Response

I used this explanation as a reference when developing the initial server and client implementation. I then compiled and executed the programs to verify that the client could connect to the server.

4. Git and GitHub

Tool-ChatGPT

Prompt / Question-Asked for help setting up Git for the project and pushing the project to GitHub.

Support Received

ChatGPT explained:

- how to initialize a Git repository;
- how to change the default branch to main;
- how to configure the repository;
- how to add files using git add;
- how to create commits;
- how to configure a GitHub remote;
- how to push the main branch.

How I Used the Response

I initialized Git for the NetMessenger project and created the initial commit.

The repository was connected to:

DMFernando0315/IT23629844-NetMessenger

The project was successfully pushed to GitHub using the main branch.

5. TCP Message Framing

Tool-ChatGPT

Prompt / Question-Asked why recv() should not be assumed to return exactly one complete command.

Support Received

ChatGPT explained that TCP is a **byte-stream protocol**, meaning:

- one command may arrive in multiple parts;
- multiple commands may arrive together;

- the application needs to identify complete newline-terminated commands.

How I Used the Response

I used this explanation while reviewing the client/server communication logic and ensured that text commands were handled using newline-based command processing.

6. Client Disconnect Handling

Tool

ChatGPT

Prompt / Question

Asked what the server should do when a client disconnects.

Support Received

ChatGPT explained that the server should:

- detect the disconnect;
- close the socket;
- remove the user from the active client state;
- clean up associated state;
- allow the connection slot to be reused;
- notify other clients when appropriate.

How I Used the Response

I reviewed the disconnect handling in my server and tested the QUIT command and client reconnection.

7. User Registration

Tool-ChatGPT

Prompt / Question-Asked how to implement username registration and personalized server responses.

Support Received

ChatGPT explained:

- username registration;
- duplicate username checking;
- maintaining connected-user information;

- returning NID-tagged responses.

How I Used the Response

I tested registration using different usernames including:

- dulara
- kamal
- user3
- user4
- user5
- user6

The application returned responses containing:

NID:6298

8. Multiple Simultaneous Clients

Tool-ChatGPT

Prompt / Question-Asked how a server can handle multiple clients simultaneously.

Support Received-ChatGPT explained approaches for concurrent client handling and the need to allow multiple client connections.

How I Used the Response

I implemented a multi-client server and tested it with more than five clients connected at the same time.

The clients included:

dulara

kamal

user3

user4

user5

user6

The LIST command successfully displayed connected users.

9. LIST and User Presence

Tool-ChatGPT

Prompt / Question-Asked how the server should maintain and display the currently connected users.

Support Received-ChatGPT explained that the server should maintain an active-user list and update it when users connect or disconnect.

How I Used the Response

I tested the LIST command.

The application successfully produced an output similar to:

OK USERS dulara, kamal, user3, user4, user5, user6 NID:6298

I also tested join/presence notifications when additional clients connected.

10. Broadcast Messaging

Tool-ChatGPT

Prompt / Question-Asked how broadcast messages should be delivered to connected clients.

Support Received-ChatGPT explained that a broadcast should be delivered to the other registered clients while the sender receives an appropriate server response.

How I Used the Response

I tested:

BCAST Hello everyone

with multiple clients connected.

The broadcast message was successfully delivered to the connected recipients.

11. Private Messaging

Tool-ChatGPT

Prompt / Question-Asked how to implement private messaging between two registered users and how to handle an invalid username.

Support Received

ChatGPT explained:

- identifying the destination username;
- finding the corresponding client;
- forwarding the private message;
- returning an error when the destination does not exist.

How I Used the Response

I tested private messaging using:

PMSG kamal Hello Kamal

The message was successfully received by kamal.

I also considered invalid-user error handling during testing.

12. Chat Rooms

Tool-ChatGPT

Prompt / Question-Asked how to manage chat-room membership and room messages.

Support Received-ChatGPT explained the need to manage:

- room creation/joining;
- room membership;
- room listing;
- room messages;
- leaving a room;
- room membership validation.

How I Used the Response

I tested the following commands:

JOIN

ROOMS

RMSG

LEAVE

The room functionality worked successfully during testing.

13. File Transfer

Tool-ChatGPT

Prompt / Question-Asked how the SENDFILE command should transfer a file through TCP.

Support Received-ChatGPT explained that the file-transfer operation needs to distinguish the text command/header from the raw file bytes and handle the declared file size correctly.

How I Used the Response

I created a test file:

test.txt

containing a small IE3010 test message.

I then used the NetMessenger client to send the file from dulara to kamal.

The recipient successfully received the file.

14. File Storage

Tool-ChatGPT

Prompt / Question-Asked how to verify the required personalized storage directory.

Support Received-The required storage structure was identified as:

./storage/IT23629844/<sender_username>/<filename>

How I Used the Response

I checked the server's storage directory after performing a file transfer and verified that transferred files were stored under the personalized registration directory.

15. Error Handling

Tool-ChatGPT

Prompt / Question-Asked how to test invalid commands and invalid destinations.

Support Received-ChatGPT suggested testing cases such as:

- private message to an unknown user;
- room message to an invalid room;
- unknown commands;
- invalid command parameters.

How I Used the Response

I used invalid-command scenarios to check that the server produced error responses instead of crashing or terminating the connection unexpectedly.

16. Server-Side Logging

Tool-ChatGPT

Prompt / Question-Asked what information should be recorded in the server log.

Support Received-ChatGPT explained that important runtime activities should be logged, including:

- client connections;

- registrations;
- commands;
- file-transfer activity;
- errors;
- disconnections.

How I Used the Response

I checked:

netmsg_IT23629844.log

after running the application and used the log as evidence of server-side activity.

17. Testing with Multiple Clients

Tool-ChatGPT

Prompt / Question-Asked for a practical test sequence for the completed application.

Support Received-ChatGPT suggested testing the main features individually and then testing them with several simultaneous clients.

How I Used the Response

I tested:

- registration;
- multiple simultaneous clients;
- LIST;
- BCAST;
- PMSG;
- JOIN;
- ROOMS;
- RMSG;
- LEAVE;
- SENDFILE;
- QUIT;
- reconnection;
- storage and logging.

The main functionality was successfully demonstrated during the testing process.

18. Final Build Check

Tool-ChatGPT

Prompt / Question-Asked what should be checked before final submission.

Support Received

ChatGPT recommended:

- checking the source files;
- checking the Makefile;
- performing a final compilation;
- checking the executable;
- checking the log;
- checking the storage directory;
- checking Git status;
- ensuring the personalized filenames are correct.

How I Used the Response

I compiled the project using:

```
make -f Makefile_9844
```

and confirmed that the client and server could be built and executed.

19. Documentation and Evidence

Tool-ChatGPT

Prompt / Question-Asked what evidence should be included in the implementation report.

Support Received

ChatGPT helped identify useful evidence including:

- server startup;
- port verification;
- multiple connected clients;
- LIST;
- broadcast messaging;
- private messaging;

- room functionality;
- file transfer;
- file storage;
- error responses;
- server logging;
- source-code sections;
- Git/GitHub evidence.

How I Used the Response

I captured terminal screenshots while running the application and planned to include the relevant screenshots in the Implementation Report.

Summary

ChatGPT was used mainly as a **technical reference and development support tool** during the NetMessenger project.

I used it to understand C socket programming, TCP communication, message framing, concurrent clients, registration, messaging, chat rooms, file transfer, error handling, logging, Git/GitHub, testing and documentation.

The suggestions were used as guidance, while the application was **compiled, executed and tested in my own Linux environment**.

The final project was tested using multiple client instances and the main required NetMessenger features were verified.